Cymatic Φ-NeRF: Quasi-Periodic Golden-Ratio Positional Encoding and 3D Nyquist Anti-Aliasing in Neural Radiance Fields

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Abstract. *Coordinate-based Multi-Layer Perceptrons (MLPs) in Neural Radiance Fields (NeRF) rely on sinusoidal positional encodings to overcome spectral bias. The standard NeRF formulation uses a dyadic geometric progression $\omega_k = 2^k\pi$ ($k = 0, ..., L–1$). We demonstrate analytically and empirically that under finite volumetric sample budgets N in $[–1, 1]^3$, dyadic scaling suffers from two distinct failure modes: (1)* ***Super-Nyquist Volumetric Aliasing****, where half of the frequency bands for $L=8$ (16π, 32π, 64π, 128π) exceed the 3D Shannon–Nyquist cutoff $\omega_{Nyq} = (\pi/2)\, N^{1/3} \approx 10.08\pi$ rad/unit (at $N = 8{,}192$), inducing severe high-frequency overfitting between sampled points; and (2)* ***Harmonic Phase Locking****, where exact integer ratios $\omega_k/\omega_j = 2^{k-j} \in Z$ cause constructive interference spikes across octaves. We introduce* ***Cymatic Φ-NeRF****, replacing dyadic 2^k scaling with powers of the Golden Ratio $\Phi = (1+\sqrt{5})/2 \approx 1.618034$ ($\log_2\Phi \approx 0.6942$ octaves/band) coupled with an optional polar cymatic standing-wave envelope $w_m(\theta) = 0.8 + 0.2|\sin(m\theta)|$. Across 3-seed held-out generalization benchmarks ($N_{train} = 8{,}192$, $N_{test} = 4{,}096$), Golden-Ratio and Cymatic Φ-encodings outperform standard dyadic NeRF by* ***+1.84 to +2.03 dB Test PSNR*** *on spherical shell fields (and* ***+5.30 dB*** *in single-seed fast convergence) and by* ***+1.00 to +1.04 dB Test PSNR*** *on a neutral 3D Schoen Gyroid minimal surface containing zero Φ-dependency in its ground truth. Explicitly matching the geometric base to the 3D Nyquist cutoff ($b_{Nyq} = \omega_{Nyq}^{1/(L-1)} \approx 1.391$) yields* ***20.76 ± 0.50 dB*** *(+4.08 dB over dyadic), confirming sub-Nyquist spectral packing as the primary physical mechanism.*

1. Introduction & Volumetric Nyquist Analysis

Coordinate MLPs map 3D spatial points $p = (x, y, z) \in [–1, 1]^3$ into Fourier features via $\gamma_b(p) = [p, \sin(b^0\pi p), \cos(b^0\pi p), ..., \sin(b^{L-1}\pi p), \cos(b^{L-1}\pi p)]$. For N uniform volumetric points in $[–R, R]^3$, the mean linear spacing per coordinate axis is $\Delta x = 2R / N^{1/3}$, yielding the 3D Shannon–Nyquist angular cutoff:

$$\omega_{Nyq} = \pi / \Delta x = (\pi N^{1/3}) / (2R) \approx 10.0794\ \pi \text{ rad/unit (for } N = 8{,}192, R = 1).$$

Proposition 1 (Sub-Nyquist Band Packing). Under $L = 8$ bands, standard dyadic scaling ($b = 2$) places 4 of 8 bands (16π, 32π, 64π, 128π) strictly above ω_{Nyq}, reaching 12.7× the Nyquist limit and causing severe inter-ray aliasing. By contrast, Golden-Ratio scaling ($b = \Phi \approx 1.6180$) places 5 of 8 bands below the Nyquist limit (a sixth, 11.09π, lies within 10% above it), while the Nyquist-matched base $b_{Nyq} = (10.0794)^{1/7} \approx 1.3911$ places all 8 of 8 bands inside the unaliased passband.

Proposition 2 (Diophantine Phase-Collision Metric). Measuring low-order integer harmonic commensurability via $C(b) = \text{mean}_{0\le j<k<L}\,[\min_{1\le m\le 8} |b^{k-j} – m| + 10^{-3}]^{-1}$ shows that dyadic octaves produce $C(2.0) = 642.88$ due to exact integer powers $2^{k-j} \in Z$, whereas $\Phi = [1; 1, 1, ...]$ gives $C(\Phi) = 2.98$ and the Nyquist-matched base $C(1.3911) = 5.84$. Because the best-performing base has the higher collision index, the gains are attributed primarily to sub-Nyquist band placement. *(Erratum v1.0.1: v1.0.0 stated 14.23 / 30.15 and 6 of 8 sub-Nyquist Φ bands.)*

2. Multi-Scene Held-Out Generalization Benchmarks

Table 1 summarizes bandwidth diagnostics and Table 2 reports 3-seed held-out test generalization (seeds 42, 43, 44; $N_{train} = 8{,}192$, $N_{test} = 4{,}096$; 300 Adam steps, hidden_dim = 128) across Scene 1 (Spherical Shell) and Scene 2 (Neutral 3D Schoen Gyroid minimal surface $\sin(3\pi x)\cos(3\pi y) + \sin(3\pi y)\cos(3\pi z) + \sin(3\pi z)\cos(3\pi x)$, which contains zero Φ-dependency).

Table 1: 3D Nyquist Bandwidth & Diophantine Diagnostics (L = 8, N = 8,192)

Encoding Variant	Base b	log2(b)	Max Band (rad)	Sub-Nyquist Bands	Collision C(b)
Standard Dyadic (2^k)	2.0000	1.0000	128.00 pi	4 / 8 (50%)	642.86

Pure Golden Ratio (Phi^k)	1.6180	0.6942	29.03 pi	5 / 8 (62.5%)	2.98
Cymatic Phi (Phi^k + w_4)	1.6180	0.6942	29.03 pi	5 / 8 (62.5%)	2.98
Nyquist-Matched (1.391^k)	1.3911	0.4762	10.08 pi	8 / 8 (100%)	5.84

Table 2: 3-Seed Held-Out Generalization Benchmark (N_train = 8,192, N_test = 4,096, 300 Steps)

Scene & Encoding Variant	Train PSNR (dB)	Test PSNR (dB)	Test Loss	Gain vs. Dyadic
Scene 1: Spherical Shell (Phi-Spiral)				
Standard Dyadic (2^k)	19.57 +/- 0.72	16.68 +/- 0.74	0.04847	Baseline
Pure Golden Ratio (Phi^k, m=0)	20.96 +/- 0.67	18.71 +/- 0.28	0.03034	+2.03 dB
Cymatic Phi-NeRF (Phi^k, m=4)	20.86 +/- 1.04	18.52 +/- 0.38	0.03087	+1.84 dB
Nyquist-Matched (1.391^k)	22.74 +/- 0.86	20.76 +/- 0.50	0.01596	+4.08 dB
Scene 2: Neutral 3D Schoen Gyroid				
Standard Dyadic (2^k)	12.49 +/- 0.60	11.99 +/- 0.06	0.47195	Baseline
Pure Golden Ratio (Phi^k, m=0)	13.61 +/- 1.54	13.03 +/- 0.95	0.44478	+1.04 dB
Cymatic Phi-NeRF (Phi^k, m=4)	13.45 +/- 1.35	12.99 +/- 0.92	0.44498	+1.00 dB
Nyquist-Matched (1.391^k)	15.24 +/- 2.65	14.74 +/- 2.24	0.35356	+2.75 dB

3. Conclusion & References

Replacing dyadic 2^k scaling with quasi-periodic Golden-Ratio (Φ^k) or 3D Nyquist-matched geometric progressions substantially suppresses inter-ray overfitting on unseen 3D coordinates across both spherical and neutral Cartesian minimal-surface geometries.

1. Mildenhall et al. (2020), *NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis*, ECCV 2020.
2. Tancik et al. (2020), *Fourier Features Let Networks Learn High Frequency Functions in Low Dimensional Domains*, NeurIPS 2020.
3. Barron et al. (2021), *Mip-NeRF: A Multiscale Representation for Anti-Aliasing Neural Radiance Fields*, ICCV 2021.